

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

- Water manometer, Part Number ST 1111 3, or equivalent
- Blowby measuring tool, Part Number 3822567, or equivalent

Additional Service Items

- No additional service items required.

General Information

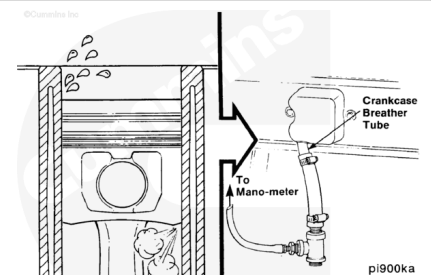
Note : Some illustrations in this procedure do not show actual engine configurations. The procedure, however, is the same.

Excessive crankcase blowby can indicate an engine or engine related component malfunction that allows combustion gases or air to enter the crankcase. This results in the buildup of higher than normal crankcase pressure, which results in increased levels of blowby.

This procedure describes how to measure crankcase blowby and how to determine which component is malfunctioning.

Blowby is typically measured for the following situations:

- Verifying engine break-in after an engine rebuild
- Troubleshooting for excessive lubricating oil out of the crankcase breather tube, commonly referred to as oil carryover (for open crankcase ventilation systems)
- Troubleshooting oil in the air intake system (for closed crankcase ventilation systems)
- Troubleshooting high crankcase pressure (for engines equipped with a crankcase pressure sensor)
- Troubleshooting possible internal engine damage (worn piston rings, valve stem seals or guides, and so forth).



This procedure provides general guidelines for measuring blowby relative to the above situations.

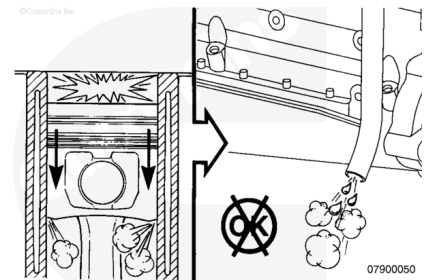
It is important to note that the terms blowby and carryover (oil out of the breather tube) are commonly used interchangeably.

When measuring blowby, if there is an excessive amount of oil coming out of the breather tube, the quantity of oil can affect the blowby measurement.

The blowby measurement is affected by the oil collecting on the orifice of the blowby measurement service tool. This reduces the size of the orifice, which results in higher than actual blowby measurements.

If this occurs, it will be necessary to:

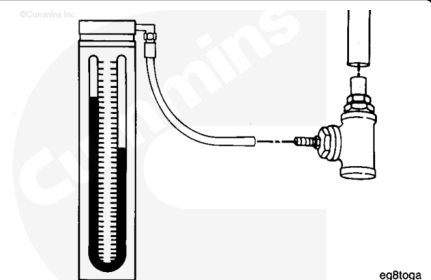
- Find a different location on the engine to measure blowby (oil fill, oil fill cap, unused turbocharger drain location, and so forth).
- Clean any oil residue from the breather and dry thoroughly before measuring blowby.
- Determine if there is an issue causing the breather to be flooded with oil, for example:
 - Incorrect oil level
 - Vehicle operation (excessive angularity, excessive engine side to side movement)
 - Internal engine components are deflecting oil toward the breather cavity (piston cooling nozzles, accessory oil drains, and so forth).
- Determine if another breather option is available for the engine being serviced.



The tools used to measure blowby are similar in design. The difference between the tools is in the size of the orifice. Different size orifices are available to more accurately measure blowby and to accommodate the wide variety of engine configurations and ratings. This is due to the fact that engine blowby is dependent on the volume of intake airflow.

For example:

- If measuring blowby on two identically configured and sized engines, but the horsepower ratings and rated speeds are different, the maximum blowby value measured will be different.



The engine with the higher horsepower rating and rated speed will have a higher volume of intake airflow, which will result in higher blowby. This means that if the smaller orifice blowby tool was used on the engine with a higher horsepower rating and rated speed, the measurement can exceed the limits of the pressure measuring tool.

Heavy Duty 15L Blowby Tool Part Number	Orifice Size
3822567	8.99 mm [0.354 in]

To measure the crankcase blowby pressure, connect a water manometer, Part Number ST 1111 3, or equivalent, pressure gauge, or transducer to the blowby measurement service tool.

Note : Water manometer, Part Number ST 1111 3, or equivalent, can measure a maximum of 944 mm [36 in] of water.

The following chart shows the relationship of measured pressure to flow rate for a 8.99-mm [0.354-in] orifice.

Blowby Conversion Table with 8.99 mm [0.354 in] Orifice and Blowby	
mm H₂O [in H₂O]	Liter [cfm] per Minute
25.4 [1]	70 [1.766]
50.8 [2]	100 [3.531]
76.2 [3]	120 [4.238]
101.6 [4]	140 [4.944]
127 [5]	158 [5.580]
152.4 [6]	172 [6.074]
177.8 [7]	185 [6.533]
203.2 [8]	198 [6.992]
228.6 [9]	210 [7.416]
254 [10]	220 [7.769]
279.4 [11]	231 [8.158]
304.8 [12]	241 [8.511]
330.2 [13]	252 [8.899]
355.6 [14]	260 [9.182]
381 [15]	270 [9.535]
406.4 [16]	280 [9.888]
431.8 [17]	289 [10.206]
457.2 [18]	298 [10.524]
482.6 [19]	305 [10.771]

Blowby Conversion Table with 8.99 mm [0.354 in] Orifice and Blowby

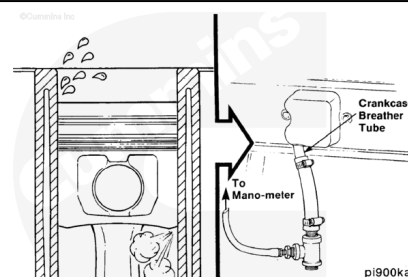
mm H ₂ O [in H ₂ O]	Liter [cfm] per Minute
508 [20]	312 [11.018]
533.4 [21]	320 [11.301]
558.8 [22]	328 [11.583]
584.2 [23]	335 [11.830]
609.6 [24]	341 [12.042]

Note : If internal engine damage is suspected to be the cause of the excessive blowby condition, other steps can be taken to confirm this.

Measuring blowby **must only** be considered when confirming engine break-in after a rebuild, or if another symptom is present. These symptoms can include:

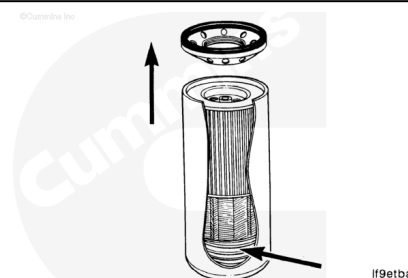
- Excessive carryover (oil out of the crankcase breather tube)
- High crankcase pressure (for engines equipped with a crankcase pressure sensor)
- Low power
- Oil consumption
- Exhaust smoke.

If no other symptom is present, blowby measurements need **not** be taken.



If internal engine damage is suspected to be the cause of the excessive blowby condition, other steps can be taken to confirm this. The steps include:

- Confirm engine maintenance practices
- Cut the oil filter open and check for debris
- Take an oil sample and inspect for contamination. Refer to Procedure 007-083 in Section 7.
(</qs3/pubsys2/xml/en/procedures/377/377-007-083.html>)



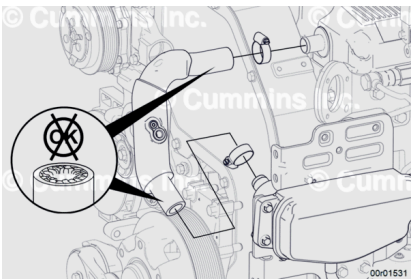
Initial Check

Note : The location and type of crankcase breathers vary by engine configuration (front gear train or rear

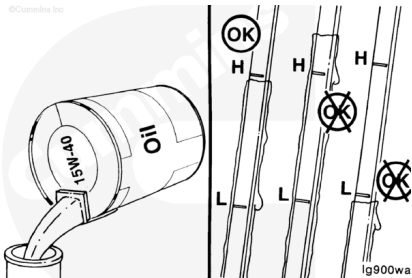
gear train) and/or engine application.

Prior to measuring blowby pressure, check the crankcase breather tube for obstructions. Refer to Procedure 003-001 in Section 3. (/qs3/pubsys2/xml/en/procedures/377/377-003-001.html)

If troubleshooting a complaint of excessive oil out of the breather tube, it can be necessary to remove the breather components to clean and remove any lubricating oil buildup before performing any blowby measurements.



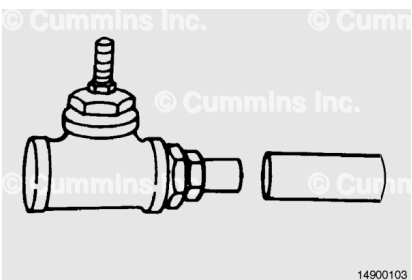
Check the engine oil level and, if necessary, proper calibration of the dipstick. If the level is too high, it can cause a higher than normal blowby pressure and/or excessive carryover. Refer to Procedure 007-009 in Section 7. (/qs3/pubsys2/xml/en/procedures/377/377-007-009-tr.html)



Measure

Obtain the appropriate blowby measurement service tool.

Heavy Duty 15L Blowby Tool Part Number	Orifice Size
3822567	8.99 mm [0.354 in]

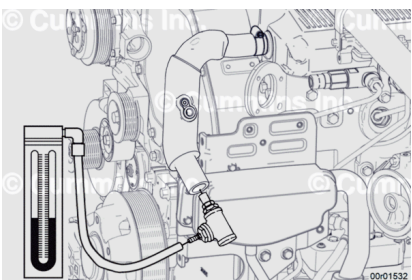


To measure the crankcase blowby pressure, connect a water manometer, Part Number ST 1111 3, pressure gauge, or transducer to the blowby measurement service tool.

Note : The location of the crankcase breather tube can vary by engine configuration and/or application.

Install the appropriate blowby service tool(s):

- For externally mounted crankcase breather systems, disconnect the tube from the inlet side of the crankcase breather. Connect the appropriate blowby service tool to the tube previously connected to the inlet side of the

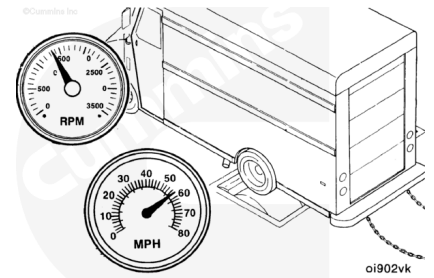


crankcase breather. Connect a water manometer pressure gauge or transducer to the blowby service tool.

Check the following for engine blowby contribution.

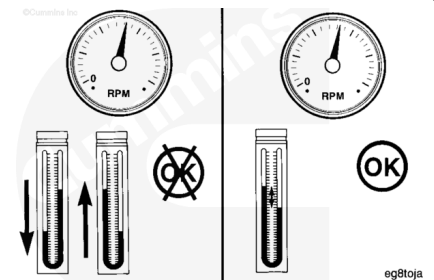
Operate the engine at rated speed and under load by either:

- For engine run-in, on a chassis dynamometer
- For engine testing, on a chassis dynamometer
- A stall speed test (for engines equipped with automatic transmissions **only**).



⚠ CAUTION ⚠

When measuring blowby and there is an excessive amount of oil coming out of the breather tube, the quantity of oil can affect the blowby measurement.



For specific instructions for the appropriate engine test, reference the appropriate procedure number listed below.

- Refer to Procedure 014-002 in Section 14.
(/qs3/pubsys2/xml/en/procedures/377/377-014-002.html)
- Refer to Procedure 014-003 in Section 14.
(/qs3/pubsys2/xml/en/procedures/377/377-014-003.html)
- Refer to Procedure 014-004 in Section 14.
(/qs3/pubsys2/xml/en/procedures/377/377-014-004.html)
- Refer to Procedure 014-008 in Section 14.
(/qs3/pubsys2/xml/en/procedures/101/101-014-008.html)
- Refer to Procedure 014-009 in Section 14.
(/qs3/pubsys2/xml/en/procedures/101/101-014-009.html)

Operate the engine at rated rpm and full load until a steady reading is obtained.

Note : When measuring blowby, the value can “spike” initially as the engine reaches peak power and rated speed. Wait for the blowby measurement to stabilize before taking a reading.

Note : For engine run-in, if a sudden increase in blowby occurs, or if blowby exceeds the maximum allowable limit during any run-in step, return to the previous step and continue the run-in. If blowby does not reach an

acceptable level, discontinue the run-in and determine the cause.

Record the steady blowby measurement.

Remove the engine blowby service tool and water manometer or pressure gauge, if the blowby is within specification.

Use the following procedure for appropriate blowby specifications. Refer to Procedure 018-023 in Section V.
(/qs3/pubsys2/xml/en/procedures/377/377-018-023.html)